

EVERY n POINTS IN THE PLANE CONTAIN $(m_1 + o(1))n$ POINTS WITH NO UNIT DISTANCE

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ABSTRACT. Let $f(n)$ be the largest integer such that every set of n points in the plane contains $f(n)$ points no two at distance 1 (Erdős Problem #1070). Let m_1 be the supremum of the upper densities of measurable planar sets with no two points at distance 1. Larman and Rogers showed $f(n) \geq m_1 n$. We prove the matching asymptotic upper bound

$$f(n) = (m_1 + o(1))n,$$

which is the first half of Conjecture 1 of Dúcz and Varga. Combined with the bound $m_1 \leq 0.247$ of Ambrus, Csiszárík, Matolcsi, Varga and Zsámboki, this gives $f(n) \leq (0.247 + o(1))n$. The proof transfers an arbitrary (non-measurable) independent set to a measurable one, using the Haar rigidity theorem for wild character laws from the OpenAI preprint *The Euclidean plane is not five-colorable* (2026); that theorem has a machine-checked Lean proof, which we have rebuilt.

1. INTRODUCTION

For a finite set $P \subset \mathbb{R}^2$ let G_P be its unit-distance graph and $\alpha(P)$ its independence number. Erdős asked to estimate

$$f(n) = \min_{P \subset \mathbb{R}^2, |P|=n} \alpha(P),$$

and in particular whether $f(n) \geq n/4$. The second question was answered negatively by Dúcz and Varga [3], who constructed a finite unit-distance graph with independence ratio below $1/4$. For the first, the forum discussion of the problem proposed $f(n) = (m_1 + o(1))n$ as a resolution, and Dúcz and Varga state as their Conjecture 1 that $f(n)/n = m_1(\mathbb{R}^2) + o(1)$ and that $m_1(\mathbb{R}^2) = \delta_{\text{Croft}}$. We prove the first half.

Definition 1.1. For measurable $A \subset \mathbb{R}^2$ let

$$\bar{\delta}(A) = \limsup_{m \rightarrow \infty} \frac{|A \cap Q_m|}{|Q_m|},$$

where Q_m is either the ball $B(0, m)$ or the square $[-m, m]^2$; every argument below works for both. Let m_1 be the supremum of $\bar{\delta}(A)$ over measurable A containing no two points at distance 1, and

$$\alpha^* = \inf \left\{ \frac{\alpha(P)}{|P|} : P \subset \mathbb{R}^2 \text{ finite, nonempty} \right\}.$$

Theorem 1.2. (1) $f(n) \geq \alpha^* n$ for every n , and $f(n)/n \rightarrow \alpha^*$.

(2) $\alpha^* \geq m_1$ (Larman–Rogers [4]).

(3) $m_1 \geq \alpha^*$.

Hence $m_1 n \leq f(n) \leq m_1 n + o(n)$.

Corollary 1.3. By [1], $m_1 \leq 0.247$; by Croft [2], $m_1 \geq 0.22936$. Hence $0.22936 n \leq f(n) \leq (0.247 + o(1))n$.

Parts (1) and (2) are elementary (Sections 2 and 3). Part (3) is the content of the paper (Section 4). Its only external input is Theorem 4.1 below, quoted from [6].

Remark 1.4 (Status of the dependency). Theorem 4.1 is proved in the unreferenced preprint [6]. It is also proved in Lean 4 with Mathlib as the declaration `OAI.PlaneFiveColor.Spectral.wild_fourier` in [7], in exactly the Fourier form used here. We rebuilt that development (Lean v4.34.1, Mathlib d13f23b); `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. A Lean formalization of the present argument on top of it is in progress at <https://github.com/daizisheng/erdos-1070>.

2. CONVERGENCE OF $f(n)/n$

Proof of Theorem 1.2(1). If $|P| = n$ then $\alpha(P) \geq \alpha^*|P|$, so $f(n) \geq \alpha^*n$.

Fix a finite P_0 with $|P_0| = s$, $\alpha(P_0) = a$. For $N \geq s$ write $N = ks + r$ with $0 \leq r < s$. Place k translates of P_0 and r further single points so that distinct pieces are more than $\text{diam}(P_0) + 2$ apart. The union P has N points and G_P is the disjoint union of the pieces' graphs, so

$$f(N) \leq \alpha(P) = ka + r \leq \frac{N}{s}a + s.$$

Thus $\limsup_N f(N)/N \leq a/s$; taking the infimum over P_0 gives $\limsup f(N)/N \leq \alpha^*$. $\square$

3. LARMAN–ROGERS: $\alpha^* \geq m_1$

Proof of Theorem 1.2(2). Let A be measurable and unit-distance-free with $\bar{\delta}(A) = \delta$, and let $P = \{p_1, \dots, p_n\}$, $\rho = \max_j |p_j|$. For every x , $(P + x) \cap A$ is independent in $G_{P+x} \cong G_P$. By Tonelli,

$$\int_{Q_m} |(P + x) \cap A| dx = \sum_j |A \cap (Q_m + p_j)| \geq \sum_j (|A \cap Q_m| - |Q_m \setminus (Q_m + p_j)|).$$

For balls and squares, $|Q_m \setminus (Q_m + p)| \leq 2\pi m|p|$, and $|Q_m| \geq \pi m^2$. Hence

$$\frac{1}{|Q_m|} \int_{Q_m} |(P + x) \cap A| dx \geq n \frac{|A \cap Q_m|}{|Q_m|} - \frac{2n\rho}{m}.$$

Along a subsequence on which the density ratio tends to δ , the right side exceeds $n\delta - \varepsilon$, so some x has $\alpha(P) \geq |(P + x) \cap A| > n\delta - \varepsilon$. Hence $\alpha(P) \geq m_1|P|$. $\square$

4. THE TRANSFER: $m_1 \geq \alpha^*$

4.1. Setting and the cited theorem. Identify $\mathbb{R}^2$ with $\mathbb{C}$. Let $F = \overline{\mathbb{Q}} \cap \mathbb{R}$, $E = F(i)$ (the algebraic numbers) and $K = \{u \in E : u\bar{u} = 1\}$, both regarded as discrete groups. Let $D = \widehat{E_{\text{disc}}}$ be the compact dual of E , with $(ad)(z) = d(az)$ and $\psi_z(d) = d(z)$ for $a, z \in E$. The continuous characters form the Borel subgroup $C = j(\mathbb{R}^2)$, $j(\xi)(z) = e^{i\xi \cdot z}$. For $x, y \in E$, $|x - y| = 1$ iff $x - y \in K$.

Theorem 4.1 ([6], Thm. 2.3; Lean: `wild_fourier`). *If ν is a K -invariant probability measure on D with $\nu(C) = 0$, then $\int_D \psi_z d\nu = 0$ for every $z \in E \setminus \{0\}$.*

We also use the following standard lemma, whose proof in [6, Lemma 4.1] we have checked line by line (spectral theorem for E_{disc} , Bochner's theorem on $\mathbb{R}^2$, Fourier uniqueness on D).

Lemma 4.2 ([6], Lemma 4.1). *Let a standard probability space carry a measure-preserving action of E with Koopman operators T_z . Then $\mathcal{H}_c = \{v \in L^2 : \lim_{z \rightarrow 0, z \in E} \|T_z v - v\|_2 = 0\}$ is L^2 of a factor $\mathcal{F}_c$ and is the range of the spectral projection on C . On $\mathcal{H}_c$ the T_z extend to a strongly*

continuous unitary representation T_x , $x \in \mathbb{R}^2$, preserving the positive cone and the constant 1. If rotations V_u ($u \in K$) satisfy $V_u T_z V_u^{-1} = T_{uz}$, then $\mathcal{H}_c$ and its projection are rotation invariant.

4.2. An invariant random independent set with density at least α^* . Let $\Omega \subset \{0, 1\}^E$ be the set of independent sets of the unit-distance graph on E . It is closed in the compact metrizable space $\{0, 1\}^E$. Translations $s_z(\omega) = \omega - z$ and rotations $\rho_u(\omega) = u\omega$ are homeomorphisms of Ω with $s_w \rho_u = \rho_u s_{u^{-1}w}$.

Lemma 4.3. *There is an $E \rtimes K$ -invariant Borel probability μ on Ω with $\mu(0 \in \omega) \geq \alpha^*$.*

Proof. One cannot simply average a single configuration over Følner sets of $E \rtimes K$: the visit counts are then non-uniform and one obtains a *weighted* independence ratio, which may be smaller than α^* . We average in two stages.

Translations. E is countable abelian, so it has Følner sets F_N . Each F_N is a finite planar set; pick a maximum independent $J_N \subset F_N$, so $|J_N| \geq \alpha^* |F_N|$. Let $\nu_N = |F_N|^{-1} \sum_{z \in F_N} \delta_{J_N - z}$. Then $\nu_N(0 \in \omega) = |J_N|/|F_N| \geq \alpha^*$ and $\|(s_w)_* \nu_N - \nu_N\|_{TV} \leq |F_N \Delta (F_N + w)|/|F_N| \rightarrow 0$. A weak-* limit point μ_0 is E -invariant, and since $\{0 \in \omega\}$ is clopen, $\mu_0(0 \in \omega) \geq \alpha^*$.

Rotations. With Følner sets $G_M \subset K$, let $\mu_M = |G_M|^{-1} \sum_{u \in G_M} (\rho_u)_* \mu_0$. Each $(\rho_u)_* \mu_0$ is E -invariant because $(s_w)_* (\rho_u)_* = (\rho_u)_* (s_{u^{-1}w})_*$, and $\mu_M(0 \in \omega) = \mu_0(0 \in \omega)$ because $0 \in u\omega \iff 0 \in \omega$. A weak-* limit μ is K - and E -invariant with $\mu(0 \in \omega) \geq \alpha^*$. $\square$

4.3. Projection to the continuous factor. On $L^2(\Omega, \mu)$ put $T_z F(\omega) = F(\omega - z)$ and $V_u F(\omega) = F(u^{-1}\omega)$; these are unitary, $V_u T_z V_u^{-1} = T_{uz}$, and $f = \mathbf{1}[0 \in \omega]$ satisfies $T_z f = \mathbf{1}[z \in \omega]$, $V_u f = f$. Since ω is independent, $f \cdot T_u f = 0$ for $u \in K$, so

$$(1) \quad \langle f, T_u f \rangle = 0 \quad (u \in K).$$

Let $p = \mathbb{E}(f \mid \mathcal{F}_c)$ and $b = f - p$. Then $0 \leq p \leq 1$, $\mathbb{E}p = \mathbb{E}f \geq \alpha^*$, and $V_u p = p$, $V_u b = b$ (Lemma 4.2).

The spectral measure λ_b of b is K -invariant: $\langle b, T_{uz} b \rangle = \langle V_u^{-1} b, T_z V_u^{-1} b \rangle = \langle b, T_z b \rangle$, and Fourier coefficients determine finite measures on D . Since $b \perp \mathcal{H}_c$, $\lambda_b(C) = 0$. By Theorem 4.1 (applied to $\lambda_b/\|b\|^2$ if $b \neq 0$), $\langle b, T_z b \rangle = 0$ for $z \neq 0$. As $\mathcal{H}_c^\perp$ is T -invariant, $\langle p, T_u b \rangle = \langle b, T_u p \rangle = 0$. With (1),

$$\int p T_u p d\mu = 0 \quad (u \in K).$$

K is dense in the unit circle (e.g. $((1 - t^2) + 2ti)/(1 + t^2)$, $t \in \mathbb{Q}$) and $x \mapsto T_x p$ is continuous, so by unitarity

$$(2) \quad \int (T_x p)(T_{x+u} p) d\mu = 0 \quad (x \in \mathbb{R}^2, |u| = 1),$$

with $0 \leq T_x p \leq 1$ and $\mathbb{E}[T_x p] = \mathbb{E}p$.

4.4. A measurable field and one good sample. Since $x \mapsto T_x p$ is continuous into the separable space $L^2(\mu)$, there is a jointly measurable $H : \mathbb{R}^2 \times \Omega \rightarrow [0, 1]$ with $H(x, \cdot) = T_x p$ for x outside a null set. By Tonelli and (2), for every m ,

$$\int_{\Omega} \int_{B(0, m)} \int_{S^1} H(x, \omega) H(x + u, \omega) d\sigma(u) dx d\mu(\omega) = 0,$$

so a conull set Ω_1 of ω has all inner integrals zero. Let $A_m(\omega) = |Q_m|^{-1} \int_{Q_m} H(x, \omega) dx \in [0, 1]$; then $\mathbb{E}A_m = \mathbb{E}p \geq \alpha^*$, and by the reverse Fatou lemma $X = \limsup_m A_m$ satisfies $\mathbb{E}X \geq \alpha^*$, hence $\mu(X \geq \alpha^*) > 0$. Fix $\omega \in \Omega_1 \cap \{X \geq \alpha^*\}$ and $h = H(\cdot, \omega)$:

$$(3) \quad \int_{B(0, m)} \int_{S^1} h(x) h(x + u) d\sigma dx = 0 \quad \forall m, \quad \limsup_m \frac{1}{|Q_m|} \int_{Q_m} h \geq \alpha^*.$$

4.5. Density points. Let $A_0 = \{h > 0\}$. By (3), $\int_{B(0,m)} \int_{S^1} \mathbf{1}_{A_0}(x) \mathbf{1}_{A_0}(x+u) d\sigma dx = 0$ for all m , and $\bar{\delta}(A_0) \geq \alpha^*$ since $h \leq \mathbf{1}_{A_0}$. Let A be the set of Lebesgue density points of A_0 ; A is Borel and $|A \triangle A_0| = 0$, so $\bar{\delta}(A) \geq \alpha^*$.

Lemma 4.4. *A contains no two points at distance 1.*

Proof. Suppose $x, y = x + u \in A$ with $|u| = 1$. For small $h_0 > 0$ let x' be uniform on $B(x, h_0)$ and, independently, u' uniform on the arc $\{e^{i\theta}u : |\theta| < h_0/2\}$. Then $\mathbb{P}(x' \notin A_0) \rightarrow 0$, and, since conditionally on u' the point $x' + u'$ is uniform on $B(x + u', h_0) \subset B(y, 2h_0)$, $\mathbb{P}(x' + u' \notin A_0) \leq 4|B(y, 2h_0) \setminus A_0|/|B(y, 2h_0)| \rightarrow 0$. So $\mathbb{P}(x' \in A_0, x' + u' \in A_0) > 0$. The law of (x', u') has bounded density with respect to $dx \otimes \sigma$ on $B(0, m) \times S^1$ for $m > |x| + 1$, contradicting (3). $\square$

Hence A is a measurable unit-distance-free set with $\bar{\delta}(A) \geq \alpha^*$, i.e. $m_1 \geq \alpha^*$. This proves Theorem 1.2(3) and the theorem.

ACKNOWLEDGEMENTS

The negative answer to the second question of #1070 is due to Dúcz and Varga [3]. The rigidity theorem and the transfer template are from [6]; the present argument is a single-label version of its Proposition 4.2.

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